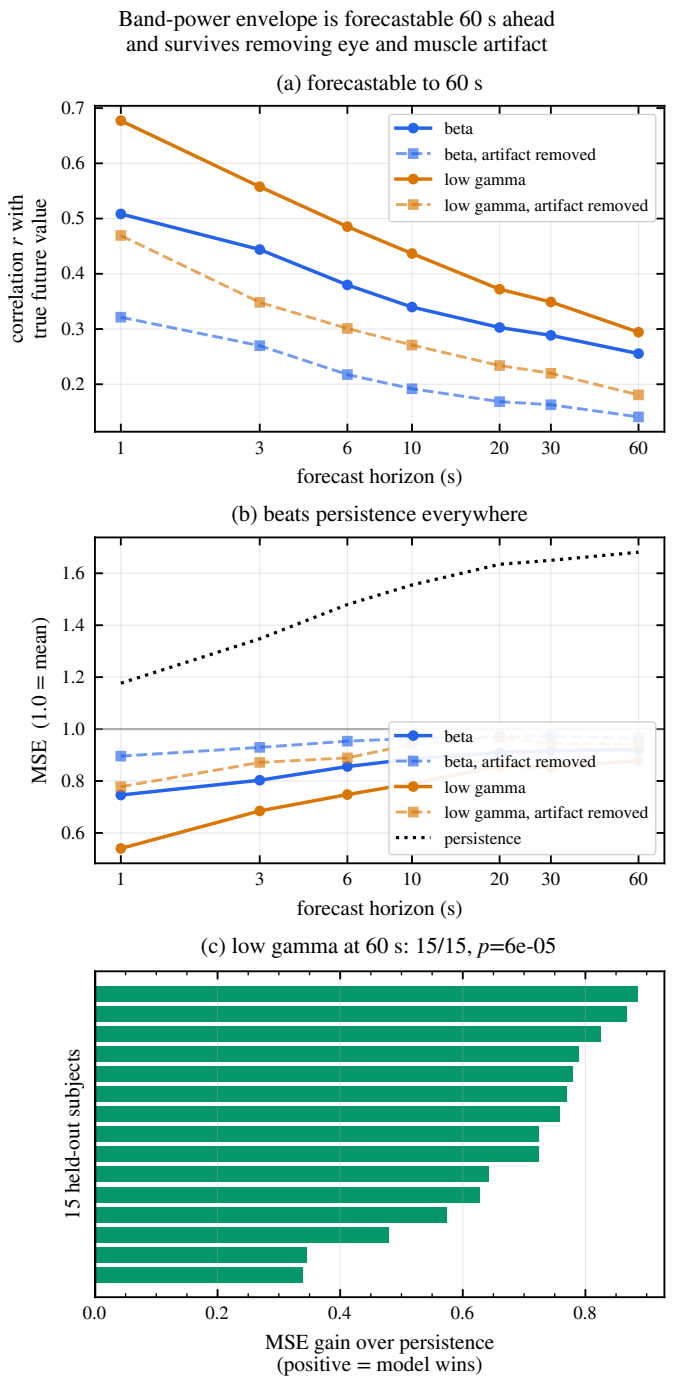
Band-power envelope is forecastable 60 s ahead
and survives removing eye and muscle artifact
(a) forecastable to 60 s
correlation r with
true future value
0.7
0.6
0.5
0.4
0.3
0.2
beta
beta, artifact removed
low gamma
low gamma, artifact removed
1
3
6
10
20
30
60
forecast horizon (s)
(b) beats persistence everywhere
MSE (1.0 = mean)
1.6
1.4
1.2
1.0
0.8
0.6
beta
beta, artifact removed
low gamma
low gamma, artifact removed
persistence
1
3
6
10
20
30
60
forecast horizon (s)
(c) low gamma at 60 s: 15/15, p=6e-05
15 held-out subjects
0.0
0.2
0.4
0.6
0.8
MSE gain over persistence
(positive = model wins)